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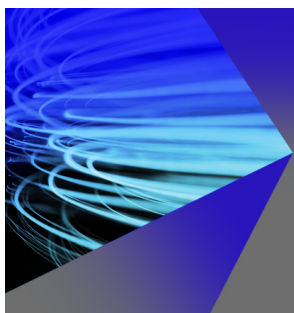
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Brillouin Scattering in Water: The Landau-Placzek Ratio*

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(Received 18 January 1967)

An experimental study of the ratio of intensities of the central line, in the molecular scattering of light, to the Brillouin components has been made in water over a temperature range from 0° to 50°C. A 75-mW He-Ne gas laser was used as a light source and the scattered radiation was analyzed by a pressure-scanned Fabry-Perot interferometer in conjunction with a dry-ice-cooled phototube detector with recorder output. The ultrahigh-frequency sound velocity is compared with values obtained by using ultrasonic techniques. There is no evidence of relaxation effects in the frequency range studied. An analysis of the Landau-Placzek ratio is made for water considered as an ideal fluid. It is shown that the usual approximations used in deriving the ratio $(\kappa_T - \kappa_S)/\kappa_S$ are not valid for water. The actual behavior of the Landau-Placzek ratio for water as a function of temperature can be explained in part when an exact solution is used.

INTRODUCTION

Fine Structure

The presence of a fine structure in the Rayleigh line during the process of molecular scattering of light from transparent fluids or solids was first predicted by Brillouin.^{1,2} He extended the Rayleigh-Einstein³ theory of molecular scattering of light by assuming the existence of a continuous spectrum of elastic waves in the homogeneous medium. The elastic waves are of thermal origin.⁴ The change in the mass density due to the elastic waves gives rise to a proportional variation in the dielectric constant of the medium. The elastic waves effectively constitute a continuous spatial grating which provides the mechanism for Brillouin scattering. Debye's representation of thermal motion as an aggregate of elastic waves in fluids and solids was also used by Mandelstam⁵ in a different analysis of the intensity of scattered light that lead to general expressions containing the earlier results of Brillouin.

Gross⁶ in an experimental study of the Raman effect was the first to ascribe certain modified radiation, contained in the spectrum of scattered light from liquids and solids, to the interaction of light waves and elastic oscillations in the medium. He suggested that a study of the modified radiation could be used to develop a new "spectroscopic" method for the determination of the acoustic velocity in solids and liquids at ultrahigh frequencies (10^9 to 10^{12} sec⁻¹), where the thermal waves would act as an elastic spatial grating. Further spectroscopic studies by Meyer and Ramm⁷ confirmed the essential features of Gross' results.

The first attempt to use this principle to determine acoustic velocities in liquids by spectrographic analysis of fine structure was by Rao.⁸ His somewhat puzzling results at the time were clarified by similar studies carried out by Venkateswaren⁹ with an improved optical system. Debye and Sears¹⁰ extended this approach and analyzed the scattering of light from progressive ultrasonic waves *induced* in liquid media by quartz transducers.

Light scattering produced by crystals had been theoretically studied by Mandelstam¹¹ as early as 1930 with no experimental corroboration. The first measurement of acoustic velocities for both transverse and longitudinal waves in vitreous silica by Brillouin spectra was reported by Flubacher,¹² and more recently by Gammon¹³ and Benedek.¹⁴ Chiao¹⁵ with the use of an intense ruby pulsed laser beam, has studied the scattering of radiation from quartz and sapphire. His results are explained in terms of *stimulated* Brillouin scattering and the coherent generation of intense ultrahigh-frequency waves. Stimulated Brillouin scattering in liquids using high-energy pulsed laser beams has also been reported.¹⁶

For a maximum intensity in the scattered radiation to exist, for a given scattering angle, Brillouin's general solution showed that there must be a unique correlation between the wavelength of the incident optical radiation and the wavelength of the elastic waves. Specifically, a Bragg relation must be satisfied.

* R. R. S. Rao, *Nature* **139**, 885 (1937).

⁹ C. S. Venkateswaren, *Proc. Indian Acad. Sci.* **15**, 371 (1942).

¹⁰ W. P. Debye and F. Sears, *Proc. Natl. Acad. Sci.* **18**, 409 (1932).

¹¹ L. Mandelstam, G. Landsberg, and M. Leontowitsch, *Z. Physik* **60**, 334 (1930).

¹² P. Flubacher, A. J. Leadbetter, J. A. Morrison, and B. P. Stoicheff, *J. Phys. Chem. Solids* **12**, 53 (1959).

¹³ R. W. Gammon and H. Z. Cummins, *Phys. Rev. Letters* **17**, 193 (1966).

¹⁴ G. B. Benedek and K. Fritsch, *Phys. Rev.* **149**, 647 (1966).

¹⁵ R. Y. Chiao, C. H. Townes, and B. P. Stoicheff, *Phys. Rev. Letters* **12**, 592 (1964).

¹⁶ J. L. Hunter, E. F. Carome, H. D. Dardy, and J. A. Bucaro, *J. Acoust. Soc. Am.* **40**, 313 (1966).

* Research supported by a National Science Foundation Institutional Grant. Part of this work was presented at the Fall meeting of the Acoustical Society of America, 3 November 1966, Los Angeles, Calif. [*J. Acoust. Soc. Am.* **40**, 1260A (1966)].

¹ L. Brillouin, *Compt. Rend.* **158**, 1331 (1914).

² L. Brillouin, *Ann. Phys. (Paris)* **17**, 88 (1922).

³ A. Einstein, *Ann. Physik* **33**, 1275 (1910).

⁴ P. Debye, *Ann. Physik* **39**, 789 (1912).

⁵ L. Mandelstam, *J. Russ. Phys.-Chem. Soc.* **58**, 831 (1926).

⁶ E. Gross, *Nature* **126**, 201, 400, 603 (1930); **129**, 722 (1932).

⁷ E. H. L. Meyer and W. Ramm, *Physik Z.* **33**, 270 (1932).

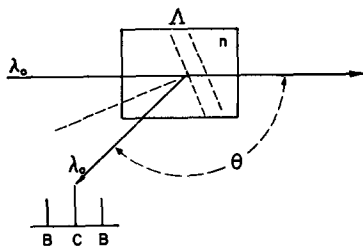


FIG. 1. Schematic representation of Brillouin scattering in liquids.

Further, it was predicted that the wavelength of the scattered radiation would be modified as a function of the optical index of refraction of the medium and the propagation velocity of the elastic waves in the media. In the scattered radiation there would be a line on either side of the central Rayleigh line forming a symmetrical pair. This effect is interpreted in terms of a Doppler shift due to the motion of the elastic waves. In liquids, where only longitudinal waves are propagated, only a single pair of lines would be observed. In solids, where longitudinal as well as one or two transverse waves can be propagated then two or more pairs of lines would be observed. The theory did not predict the existence of the central Rayleigh component. For the case of a liquid, the scattering process is shown in Fig. 1. The equations of interest in Brillouin's general scattering equation are

$$\Lambda = \lambda_0 / (2n \sin \frac{1}{2}\theta), \quad (1)$$

$$\Delta\lambda/\lambda_0 = \pm 2n(V/c) \sin \frac{1}{2}\theta, \quad (2)$$

$$f_s = 2n(V/\lambda_0) \sin \frac{1}{2}\theta. \quad (3)$$

Equation (1) gives the elastic, or acoustic, wavelength of the continuum of elastic waves present in the medium that will be effective in the scattering process at a given scattering angle. The sound velocity can be calculated from Eq. (2) by measuring the Doppler shift about the central component. The corresponding ultrahigh frequency of the acoustic waves is determined from Eq. (3).

Line Shapes

In Fig. 2 is a representation of the line shapes, and their superposition, in the scattered radiation where the central component is the Rayleigh line. The line shapes are assumed to be Lorentzian. As will be seen below the central line is due to scattering from non-propagating local entropy fluctuations in the liquid and the Brillouin components are due to scattering from adiabatic pressure fluctuations which propagate as acoustic or elastic waves. For the classical or ideal liquid (i.e., a nondispersive liquid), the line shape of the central component is also related to the coefficient of thermal conductivity while the line shapes of the Brillouin components are associated with the coefficient of bulk viscosity. Due to instrumental broadening it is

often difficult to experimentally detect a significant difference in the line shapes of these components when the coefficient of bulk viscosity is small. Optical heterodyne techniques have recently been developed to study the natural line shape of the central component in fluids near their critical point¹⁷ as well as in normal liquids.¹⁸

The ratio of intensities of the Rayleigh component (I_c) to the Brillouin components ($2I_B$) should be a constant for an ideal fluid at a given temperature and related to the thermodynamic properties of the liquid. An experimental measurement of the intensity ratio using an He-Ne laser as the exciting light source was first reported by Rank.¹⁹ Their results showed a positive departure from the predicted ratio in the behavior of real liquids. This effect could not be ascribed to parasitic scattering in the optical cell nor to depolarization effects in the liquid.

Mountain^{20,21} in a recent study analyzed the scattered radiation associated with a frequency spectrum of density fluctuations in a dispersive fluid, where the internal vibrational degrees of freedom are weakly coupled to the translational degrees of freedom in the fluid. In this analysis the thermal relaxation of the vibrational degrees of freedom is described by a single relaxation time.

The effect of the relaxation mechanism is to: (1) Leave the line shape of the central thermal mode unaffected, (2) to broaden the line shape of the Brillouin phonon modes, and (3) to give rise to a new non-propagating mode due to density fluctuations associated with the energy exchange between the internal vibrational modes and the translational modes. The experimental existence of this latter mode was recently established.²²

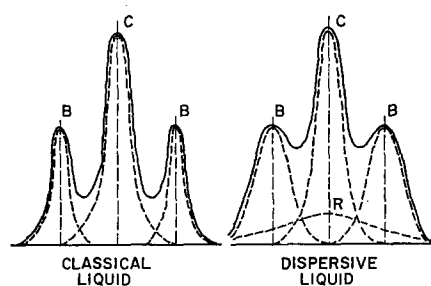


FIG. 2. Lines shapes of the fine structure in Brillouin scattering for ideal and for dispersive liquids.

¹⁷ N. C. Ford and G. B. Benedek, *Phys. Rev. Letters* **15**, 649 (1965).

¹⁸ J. B. Lastovka and G. B. Benedek, *Phys. Rev. Letters* **17**, 1039 (1966).

¹⁹ D. H. Rank, M. Kiess, U. Fink, and T. A. Wiggins, *J. Opt. Soc. Am.* **55**, 925 (1965).

²⁰ R. D. Mountain, *J. Res. Natl. Bur. Std.* **70A**, 207 (1966).

²¹ R. D. Mountain, *J. Chem. Phys.* **44**, 832 (1966).

²² W. S. Gornall, G. I. A. Stegeman, B. P. Stoicheff, R. H. Stolen, and V. Volterra, *Phys. Rev. Letters* **17**, 297 (1966).

It is possible with Mountain's theory to experimentally determine the vibrational relaxation time of a given liquid by light-scattering techniques from a measurement of: (1) velocity dispersion (displacement of the Brillouin components,²³⁻²⁶ (2) excess absorption coefficient (broadening of the Brillouin line shapes),²⁷ (3) the ratio of intensities of the Brillouin components to the central components.²⁸

The present study was concerned with the measurement of the ultrahigh-frequency velocity of sound in water by light-scattering techniques, and the determination of the ratio of intensities of the central component to the Brillouin components in the scattered radiation from water. Both measurements were made as a function of temperature.

The contribution of vibrational specific heat to the total specific heat is small over the temperature range studied, about 0.2% at room temperature. There will be no detectable broadening of the line shapes of the phonon modes. In addition, the nonpropagating coupling mode predicted by Mountain for relaxation processes would be absent. Water should behave as an ideal or classical liquid. This study was made to evaluate the assumptions made in the theoretical treatments of scattering of light from ideal and from dispersive liquids; namely, the importance of temperature fluctuations of the dielectric constant compared to the corresponding density fluctuations in these liquids.²⁹

THEORETICAL

Classical Light Scattering

The following analysis is an extension of a development recently published by Cummins and Gammon.³⁰ The intensity of scattered radiation in the classical theory of light scattering is proportional to the mean-square fluctuation of the dielectric constant $\langle(\Delta\epsilon)^2\rangle$ about its mean value in a scattering medium.

It is necessary to relate this mean-square fluctuation to fluctuations in thermodynamic quantities so chosen that they are statistically independent. This will permit cross-term products to vanish. The two classical approaches are as follows:

Einstein-Smoluchowski: $\epsilon = \epsilon(\rho, T)$

$$\langle(\Delta\epsilon)^2\rangle = (\partial\epsilon/\partial\rho)_T^2 \langle(\Delta\rho)^2\rangle + (\partial\epsilon/\partial T)_\rho^2 \langle(\Delta T)^2\rangle,$$

²³ G. B. Benedek, J. B. Lastovka, K. Fritsch, and T. Greytak, *J. Opt. Soc. Am.* **54**, 1284 (1964).

²⁴ R. Y. Chiao and B. P. Stoicheff, *J. Opt. Soc. Am.* **54**, 1286 (1964).

²⁵ P. A. Fleury and R. Y. Chiao, *J. Acoust. Soc. Am.* **39**, 751 (1966).

²⁶ J. E. Piercy and G. R. Hanes, *J. Chem. Phys.* **43**, 3400 (1965).

²⁷ G. R. Hanes, R. Turner, and J. E. Piercy, *J. Acoust. Soc. Am.* **38**, 1057 (1965).

²⁸ C. L. O'Connor and J. P. Schlupf, *J. Acoust. Soc. Am.* **40**, 663 (1966).

²⁹ D. J. Coumou, E. L. Mackor, and J. Hijmans, *Trans. Faraday Soc.* **60**, 1539 (1964).

³⁰ H. Z. Cummins and R. W. Gammon, *J. Chem. Phys.* **44**, 2785 (1966).

Landau-Placzek: $\epsilon = \epsilon(S, P)$

$$\langle(\Delta\epsilon)^2\rangle = (\partial\epsilon/\partial S)_P^2 \langle(\Delta S)^2\rangle + (\partial\epsilon/\partial P)_S^2 \langle(\Delta P)^2\rangle.$$

The choice of thermodynamic variables is such that the Einstein-Smoluchowski formulation gives an expression for the total intensity of scattered radiation {the central component plus the Brillouin components: $[(I_c + 2I_B)]$ }, while the Landau-Placzek formulation, with the choice of different thermodynamic parameters, is able to separate out the components in the scattered radiation; namely, (I_c) and $(2I_B)$. In this latter case the central Rayleigh component is due to scattering from nonpropagating local entropy fluctuations in the liquid, $(\partial\epsilon/\partial S)_P^2$. It may also be viewed as the decay of a fluctuation by a thermal diffusion process and is sometimes designated as the thermal mode. The Brillouin components are associated with scattering from adiabatic pressure fluctuations, $(\partial\epsilon/\partial P)_S^2$, which propagate as elastic waves in the scattering fluid. This latter mechanism can also be considered as scattering from the phonon modes.

From the theory of thermodynamic fluctuations,³¹ we have

$$\langle(\Delta\rho)^2\rangle \propto \rho^2 T \kappa_T / v,$$

$$\langle(\Delta T)^2\rangle \propto T^2 / c_V \rho v,$$

$$\langle(\Delta S)^2\rangle \propto c_P \rho v,$$

$$\langle(\Delta P)^2\rangle \propto T / v \kappa_S,$$

where the proportionality constant for each of the above thermodynamic expressions is the Boltzmann constant.

Using the following thermodynamic equivalents in the Landau-Placzek formulation:

$$(\partial\epsilon/\partial P)_S = (\partial\epsilon/\partial\rho)_{SP} \kappa_S,$$

$$(\partial\epsilon/\partial S)_P = (\partial\epsilon/\partial T)_P T / c_P \rho v,$$

we have for the intensity of scattered radiation

Einstein-Smoluchowski:

$$(I_c + 2I_B) \propto \rho^2 T \kappa_T \left(\frac{\partial\epsilon}{\partial\rho} \right)_T^2 + \left(\frac{T^2}{c_V \rho} \right) \left(\frac{\partial\epsilon}{\partial T} \right)_P^2 \quad (4)$$

Landau-Placzek:

$$(2I_B) + (I_c) \propto \rho^2 T \kappa_S \left(\frac{\partial\epsilon}{\partial\rho} \right)_S^2 + \left(\frac{T^2}{c_P \rho} \right) \left(\frac{\partial\epsilon}{\partial T} \right)_P^2 \quad (5)$$

The Landau-Placzek Ratio

The ratio of intensities of the central Rayleigh component to the total Brillouin components is known as the Landau-Placzek ratio. It was formulated by them without proof in a brief published in 1934³² and derived

³¹ L. D. Landau and E. M. Lifshitz, *Statistical Physics* (Addison-Wesley Publ. Co., Inc., Reading, Mass., 1958), Chap. 12.

³² L. Landau and G. Placzek, *Physik Z. Sowjetunion* **5**, 172 (1934).

in a later text.³³ Their procedure was to take the Einstein-Smoluchowski expression $(I_c + 2I_B)$, Eq. (4), and divide by the Brillouin term $(2I_B)$ from their formulation, Eq. (5),

$$(I_c + 2I_B)/(2I_B) = \frac{[\rho^2 T \kappa_T (\partial \epsilon / \partial \rho)_T^2 + T^2 / c_V \rho (\partial \epsilon / \partial T)_\rho^2]}{[\rho^2 T \kappa_S (\partial \epsilon / \partial \rho)_S^2]} \quad (6)$$

since by a Jacobian transformation,

$$(\partial \epsilon / \partial \rho)_S = (\partial \epsilon / \partial \rho)_T + T / c_V \rho^2 (\partial P / \partial T)_\rho (\partial \epsilon / \partial T)_\rho$$

and if it is assumed that the temperature fluctuations $(\partial \epsilon / \partial T)_\rho$ are small compared to the density fluctuations $(\partial \epsilon / \partial \rho)_T$, then Eq. (6) reduces to

$$I_c / 2I_B = \gamma - 1 \quad (7)$$

which is the Landau-Placzek ratio.

It should be noted that this ratio can be obtained directly from Landau-Placzek expression from the intensity of scattered radiation in Eq. (5), namely,

$$(I_c) / (2I_B) = \frac{[T^2 / c_P \rho (\partial \epsilon / \partial T)_P^2]}{[\rho^2 T \kappa_S (\partial \epsilon / \partial \rho)_S^2]} \quad (8)$$

since $(\partial \epsilon / \partial T)_P = (\partial \epsilon / \partial \rho)_T (\partial \rho / \partial T)_P + (\partial \epsilon / \partial T)_\rho$ and for $(\partial \epsilon / \partial T)_\rho \ll (\partial \epsilon / \partial \rho)_T$ then Eq. (8) reduces to

$$(I_c) / (2I_B) = (\kappa_T - \kappa_S) / \kappa_S = \gamma - 1.$$

Fabelinskii³⁴ questioned the validity of the assumption made above and pointed out that what is important is not $(\partial \epsilon / \partial T)_\rho \ll (\partial \epsilon / \partial \rho)_T$ but rather whether the second term small is compared to the first term in Eqs. (6) and (8). It does not seem that he proceeded further after making this observation. An analysis of this assumption is as follows.

Transformation of Variables

Numerical data on the variation of the dielectric properties of liquids with changes in density and temperature for isobaric and isentropic processes are not available. Using the classical relationship between the dielectric constant and the optical index of refraction, $\epsilon = n^2$ for a specified optical frequency in an isotropic medium, it is possible to transform the variables in the above scattering expressions as follows:

$$\begin{aligned} (\partial \epsilon / \partial \rho)_T &= (2n / \rho \kappa_T) (\partial n / \partial P)_T, \\ (\partial \epsilon / \partial T)_\rho &= 2n [(\partial n / \partial T)_P + (\beta / \kappa_T) (\partial n / \partial P)_T], \\ (\partial \epsilon / \partial \rho)_S &= (2n / \rho \kappa_S) [(\partial n / \partial P)_T + (\beta T / \rho c_P) (\partial n / \partial T)_P], \\ (\partial \epsilon / \partial T)_P &= 2n (\partial n / \partial T)_P. \end{aligned}$$

³³ L. D. Landau and E. M. Lifshitz, *Electrodynamics of Continuous Media* (Addison-Wesley Publ. Co., Inc., Reading, Mass., 1960).

³⁴ I. L. Fabelinskii, *Usp. Fiz. Nauk.* **63**, 355 (1957) [English transl.: *Soviet Phys.—Usp.* **63**, 474 (1957)] AEC translation 3973 Pt. 1 (1960).

Substituting into (4) and (5) above, we have

Einstein-Smoluchowski:

$$(I_c + 2I_B) \propto \left(\frac{T}{\kappa_T} \right) \left(\frac{\partial n}{\partial P} \right)_T^2 + \left(\frac{T^2}{c_V \rho} \right) \times \left[\left(\frac{\partial n}{\partial T} \right)_P + \frac{\beta}{\kappa_T} \left(\frac{\partial n}{\partial P} \right)_T \right]^2, \quad (9)$$

Landau-Placzek:

$$(I_c) + (2I_B) \propto \left(\frac{T^2}{c_P \rho} \right) \left(\frac{\partial n}{\partial T} \right)_P^2 + \left(\frac{T}{\kappa_S} \right) \times \left[\left(\frac{\partial n}{\partial P} \right)_T + \frac{\beta T}{\rho c_P} \left(\frac{\partial n}{\partial T} \right)_P \right]^2. \quad (10)$$

These expressions are in terms of changes of index of refraction with temperature and pressure and of other thermodynamic properties whose values for water are listed in Table I.

These equations are exact expressions. It is possible to determine the intensity ratio in either of two ways: combining the E-S equation $(I_c + 2I_B)$ with the $2I_B$ term of the L-P expression, or to use the (I_c) and the $(2I_B)$ terms of the L-P expression directly. A computer program was written to handle data for water over a temperature range from 0° to 10°C in 1°C intervals and from 10° to 50°C in 5°C intervals. It was thought possible to be able to distinguish between the two exact expressions as well as to determine their common difference from the approximate Landau-Placzek ratio derived from Eqs. (6) and (8).

The results of these numerical calculations are shown in Fig. 3 where the exact expression for the ratio of intensities is the same whether we use the E-S expression together with the isentropic term from the L-P formulation, or using simply the ratio of local entropy fluctuation term to the isentropic pressure fluctuation term in the L-P expression. In both cases, however,

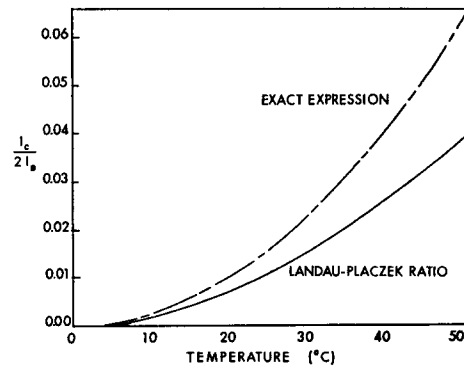


FIG. 3. Graphic representation of exact and approximate expressions for the intensity ratio $(I_c / 2I_B)$ in water as a function of temperature.

the exact expression does show a departure from the approximate Landau-Placzek ratio.

Mountain has analyzed the spectral distribution of light scattered by *temperature* fluctuations³⁵ as well as the spectral characteristics associated with *density* fluctuations where the internal vibrational degrees of

freedom are weakly coupled to the translational degrees of freedom in the fluid.²⁰

In the latter case where the thermal relaxation of the vibrational degrees of freedom is described by a single relaxation time, τ , it was assumed that $(\partial\epsilon/\partial\rho)\tau \gg (\partial\epsilon/\partial T)_\rho$. The intensity ratio becomes

$$\frac{I_c}{2I_B} = \frac{(1-1/\gamma)[(V_0/V)^4 + V^2 K^2 \tau^2] + (V_\infty^2 - V_0^2) K^2 \tau^2 - (V^2/V_0^2 - 1)[(V_0/V)^4 + V_0^2 K^2 \tau^2 (1-1/\gamma)]}{[1 - V_0^2/V^2 (1-1/\gamma)][V^2 K^2 \tau^2 + (V_0/V)^2] - (V_\infty^2 - V_0^2) K^2 \tau^2}, \quad (11)$$

where K is the acoustic wave vector related to the incident radiation vector K_0 , and the scattering angle θ by $K = 2nK_0 \sin \frac{1}{2}\theta$, V is determined from Eq. (2), V_∞ and V_0 are the limiting values of the velocity for the presence of, and absence of, relaxation of vibrational states, respectively. In the absence of a relaxation process in a fluid Eq. (11) reduces to $(\gamma-1)$, the Landau-Placzek ratio.

In the case where there is no relaxation mechanism, and only temperature and density fluctuations of the dielectric constant are considered, Mountain's analysis for the intensity ratio leads to

$$\frac{I_c}{2I_B} = \frac{(\partial\epsilon/\partial\rho)_T \rho^2 T_{KT} (1-1/\gamma) - 2(\partial\epsilon/\partial\rho)_T (\partial\epsilon/\partial T)_\rho (\rho T_{KT}/\beta) (1-1/\gamma) + (\partial\epsilon/\partial T)_\rho^2 T^2/\rho c_P}{(\partial\epsilon/\partial\rho)_T \rho^2 T_{KT}/\gamma + 2(\partial\epsilon/\partial\rho)_T (\partial\epsilon/\partial T)_\rho (\rho T_{KT}/\beta) (1-1/\gamma) + (\partial\epsilon/\partial T)_\rho^2 (T^2/\rho c_V) (1-1/\gamma)}, \quad (12)$$

which agrees exactly with the thermodynamical expression derived above. For the case of $(\partial\epsilon/\partial\rho)_T \gg (\partial\epsilon/\partial T)_\rho$ Eq. (12) reduces again to the Landau-Placzek expression.

It was pointed out that (S, P) rather than (ρ, T) appear to be the "natural" variables for discussing hydrodynamic fluctuations, except in the case where the bulk viscosity is frequency dependent. In this case S and p both develop cross terms and there is no particular advantage in considering (S, P) over (ρ, T) .

EXPERIMENTAL

The system used in the experimental studies of light scattering from water schematically in Fig. 4 and has been described earlier.²⁸ The incident beam of radiation used for scattering in liquids is from a 75-mW helium-neon gas laser operating at 6328 Å. The scattering liquid is contained in a cylindrical optical cell 10 cm long and 3 cm in diameter. Large dimensions are used to reduce the scattering contribution from the walls to the Rayleigh component. The water sample was prepared by repeated slow distillation in a closed system

that had been pumped free of air. The scattering cell was temperature controlled by an external circulating liquid-bath system. The temperature of the scattering liquid was measured to $\pm 0.1^\circ\text{C}$ by a calibrated thermistor imbedded in the scattering cell. The scattering angle was measured to $\pm 0.1^\circ$.

The improved optical arrangement for light collecting and radiation analyzing permits a very small fraction (about 10^{-3}) of the first order of the interferometer fringe pattern to be seen by the phototube.³⁶ The small effective "source" pinhole reduces to a minimum the effect of a finite aperture on the recorded line shape and over-all resolution. In addition, the pinhole is the only opening needed between the scattering region and the photomultiplier detecting system. The problem of shielding the phototube from extraneously scattered light is reduced. One of the significant advantages in the present optical arrangement is that the signal to noise ratio, with a large portion of the phototube surface being used, is increased considerably.

Figures 5 to 7 show typical recorded outputs of the spectral structure of scattered radiation from water at 50° , 20° , and 4°C , together with the gradual disappearance of the central Rayleigh component. Two measurements were made from the recorder trace: (1) The displacements of the doublets from the unshifted central line from which the acoustic velocity was determined; and (2) the integrated areas under the central line and the doublet components using procedures described elsewhere.²⁸

The velocity results are shown in Fig. 8 where the solid line is based on ultrasonic data by Greenspan and Tschiegg.³⁷ These values are tabulated in Table II.

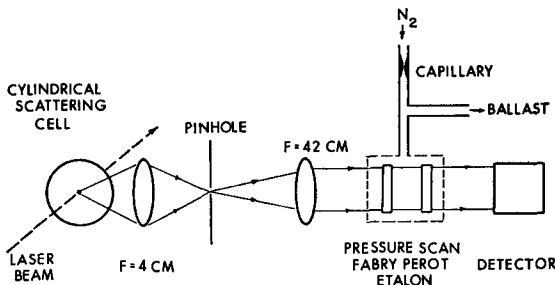


FIG. 4. Schematic of the light-scattering apparatus and the pressure-scan analyzing system. Diameter of pinhole is 0.25 mm.

³⁵ R. D. Mountain (private communication).

³⁶ J. H. Jaffe, D. H. Rank, and T. A. Wiggins, J. Opt. Soc. Am. **45**, 636 (1955).

³⁷ M. Greenspan and C. E. Tschiegg, J. Res. Natl. Bur. Std. **59**, 249 (1957).

TABLE I. Thermodynamic properties and optical properties (at 6328 a.u.) of water as a function of temperature.

T (°C)	ρ (g/cc) ^a	$V \times 10^{-6}$ (cm/sec) ^b	$\kappa_S \times 10^{11}$ (cm ² /dyn) ^c	$\kappa_T \times 10^{11}$ (cm ² /dyn) ^d	$\beta \times 10^5$ (°C ⁻¹) ^e
0	0.9998676	1.4027	5.0831	5.0869	-7.70
1	0.9999265	1.4077	5.0467	5.0484	-5.01
2	0.9999678	1.4126	5.0116	5.0123	-3.29
3	0.9999922	1.4173	4.9783	4.9785	-1.61
4	1.0000000	1.4220	4.9454	4.9454	0.00
5	0.9999919	1.4265	4.9143	4.9144	1.59
6	0.9999683	1.4309	4.8842	4.8849	3.11
7	0.9999297	1.4352	4.8552	4.8566	4.59
8	0.9998765	1.4395	4.8265	4.8289	6.03
9	0.9998092	1.4436	4.7994	4.8031	7.42
10	0.9997281	1.4476	4.7733	4.7785	8.78
15	0.9991286	1.4663	4.6551	4.6708	15.07
20	0.9982336	1.4827	4.5568	4.5868	20.66
25	0.9970751	1.4970	4.4754	4.5226	25.70
30	0.9956783	1.5094	4.4083	4.4752	30.30
35	0.9940635	1.5201	4.3535	4.4421	34.56
40	0.9922473	1.5292	4.3097	4.4217	38.50
45	0.99024	1.5367	4.2764	4.4127	42.11
50	0.98807	1.5429	4.2514	4.4114	45.64

T (°C)	C_P/R^f	$C'/R \times 10^3$ ^g	$(\gamma-1) \times 10^{10}$ ^h	n^i	$-(\partial n/\partial T)_P \times 10^6$ ⁱ (°C ⁻¹)	$(\partial n/\partial P)_T \times 10^{11}$ ⁱ (cm ² /dyn)
0	9.1381	1.5847	0.75	1.33263	1.24	1.66
1	9.1296	1.6221	0.32	1.33261	5.23	1.65
2	9.1220	1.6601	0.14	1.33260	9.21	1.63
3	9.1158	1.6987	0.03	1.33259	13.19	1.62
4	9.1099	1.7379	0.00	1.33257	17.17	1.61
5	9.1045	1.7776	0.03	1.33255	21.15	1.60
6	9.0993	1.8178	0.13	1.33253	25.11	1.59
7	9.0945	1.8587	0.28	1.33250	29.07	1.58
8	9.0904	1.9001	0.50	1.33247	33.03	1.57
9	9.0863	1.9421	0.77	1.33244	36.97	1.56
10	9.0828	1.9847	1.09	1.33240	40.90	1.56
15	9.0690	2.2061	3.36	1.33214	60.36	1.52
20	9.0605	2.4420	6.57	1.33180	79.36	1.50
25	9.0555	2.6925	10.55	1.33135	97.77	1.47
30	9.0531	2.9576	15.17	1.33082	115.4	1.45
35	9.0525	3.2373	20.35	1.33020	132.2	1.42
40	9.0534	3.5314	25.97	1.32950	147.9	1.40
45	9.0553	3.8400	31.87	1.32872	162.4	1.38
50	9.0579	4.1692	38.32	1.32788	175.5	1.36

^a American Institute of Physics Handbook, D. E. Gray, Ed. (McGraw-Hill Book Co., New York, 1963), 2nd ed. Sec. 2-147.

^b Reference 37.

^c $\kappa_S = 1/\rho V^2$.

^d $\kappa_T = 1/\rho(1/V^2 + \beta^2 T/c_P)$.

^e $\beta = -1/\rho(\partial \rho/\partial T)_P$.

^f Reference a, Sec. 4-52.

^g $C'/R = \sum u^2 e^u (e^u - 1)^{-2}$; $u = 1.4390 \omega/T$; for water; $\omega_1 = 3651.7(1)$, $\omega_2 = 1595.0(1)$, $\omega_3 = 3755.8(1)$.

^h $\gamma = \kappa_T/\kappa_S$.

ⁱ International Critical Tables, E. W. Washburn, Ed. (McGraw-Hill Book Co., New York, 1930), Vol. 7, p. 13.

TABLE II. Experimental values of the sound velocity and the ratio of intensities ($I_c/2I_B$) in water at 4.5 Gc/sec as a function of temperature.

T (°C)	V (m/sec) exptl	V (m/sec) ^a	$(I_c/2I_B) \times 10^3$ exptl	$(I_c/2I_B) \times 10^3$ corr	$(I_c/2I_B) \times 10^3$ ^b	$(I_c/2I_B) \times 10^3$ ^c
4.0	1420	1422	8±3	0	0	0
5.6	1427	1429	...	...	...	...
8.0	1436	1439	11	3	1	0
10.2	1445	1448	...	...	...	...
12.8	1458	1458	...	...	...	...
15.7	1466	1468	18	10	5	3
19.7	1484	1482	21	13	9	6
19.9	1481	1482	...	...	...	...
25.0	1496	1497	25	17	15	10
30.1	1509	1509	34	26	21	15
40.0	1527	1529	40	32	39	26
50.0	1543	1542	57	49	62	38

^a Reference 37,

^b Equation (9),

^c Equation (7),

FIG. 5. Spectral structure of scattered radiation from water at 50°C.

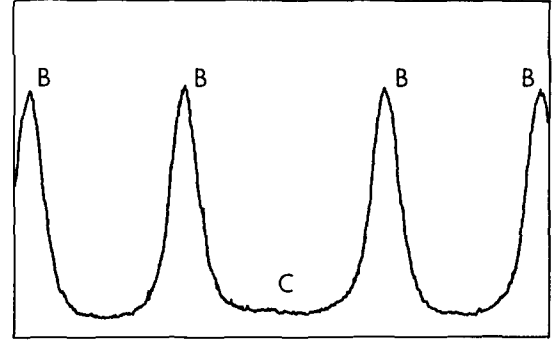
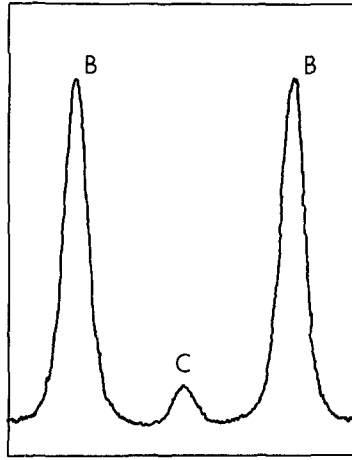


FIG. 7. Spectral structure of scattered radiation from water at 4°C.

Agreement is to better than one part in 700, which is the order of magnitude of the uncertainty in the scattering angle.

In Fig. 9 the experimental measurements of the ratio of intensities of the Rayleigh component to the Brillouin component as a function of the temperature are shown. The solid points are the original experimental data. If it is assumed that at 4°C the ratio of intensities should approach zero, on the basis of both exact and approximate solutions, we can then ascribe the intensity in the central component at this temperature to parasitical scattering in the system. The open points represent the corrected experimental data.

CONCLUSIONS

In general, it appears that the exact expression for the ratio of intensities gives better agreement with the experimental data for water than the approximate Landau-Placzek ratio. The poorer agreement at higher temperatures may be due in part to the necessity of

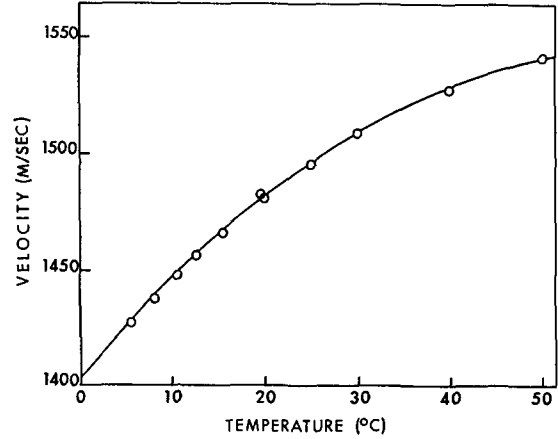


FIG. 8. Plot of experimental values of sound velocity in water at 4.5 Gc/sec as a function of temperature. Solid line is experimental data from ultrasonic measurements.

FIG. 6. Spectral structure of scattered radiation from water at 20°C.

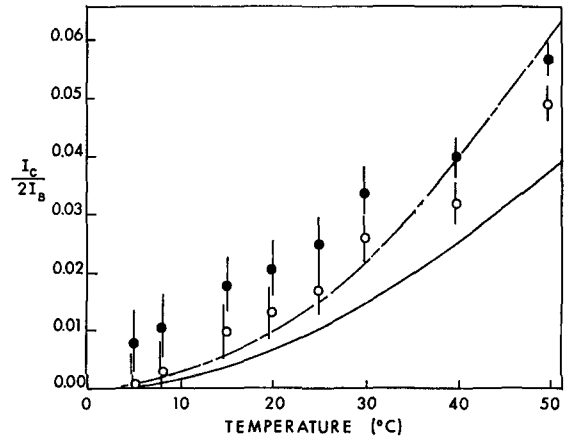
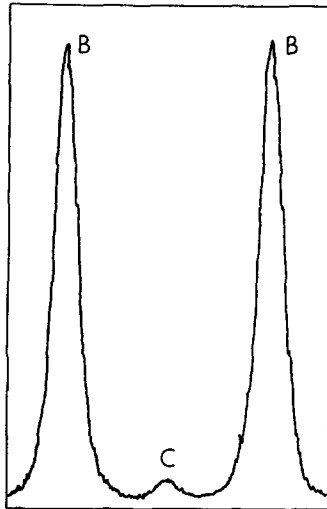


FIG. 9. The ratio of intensities $I_C/2I_B$ in water at 4.5 Gc/sec as a function of temperature. The solid line is the Landau-Placzek ratio. The dashed line is from the exact solution for the intensity ratio.

using extrapolated data for the pressure dependence of the refractive index. It should be mentioned that measurements of the ratio of intensities is a particularly difficult experimental problem, especially with water which is a notoriously "noisy" liquid.

In particular, to determine relaxation effects in dispersive liquids, velocity dispersion measurements and excess absorption measurements from Brillouin scattering are to be preferred to the ratio of intensity measurements. This latter quantity, aside from evident experimental problems, is quite insensitive to variations in relaxation times. This approach can provide the proper order of magnitude of a relaxation time, but there is a considerable range of uncertainty in the numerical coefficient.

What is understood is the mechanism by which the experimental value of the Landau-Placzek ratio will show values in excess of the predicted value for dispersive liquids; namely, relaxation effects that broadened the Brillouin components and introduce the new mode of scattered radiation predicted by Mountain. For ideal nondispersive liquids an exact solution for the Landau-Placzek ratio will better explain the behavior of a classical liquid.

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Thermodynamic Properties of Octafluoropropane from 14°K to Its Normal Boiling Point. An Estimate of the Barrier to Internal Rotation from the Entropy and Heat Capacity of the Gas

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The heat capacity of octafluoropropane has been measured from 14°K to its normal boiling point of 236.42°K for a sample of greater than 99.99 mole % purity. The following heats of transition and transition temperatures were found: Crystal I-Crystal II, 849.8 cal/mole, 99.37°K; Crystal I-liquid, 114.1 cal/mole, 125.46°K; and liquid-vapor (1 atm), 4723 cal/mole, 236.42°K. The vapor pressure up to 1 atm pressure is represented by the equation

$$\log_{10} P_{\text{mm}} = 20.85981 - (1584.677/T) - 4.750509 \log_{10} T,$$

and the density of the liquid between 197° and 234°K by the equation

$$d(\text{g/ml}) = 2.471 - 0.003628T.$$

The value of the entropy of the ideal gas at 1 atm pressure at the normal boiling point calculated from the calorimetric data is 88.75 cal/(mole·°K).

An average potential barrier of 3300 cal/mole restricting the rotation of the trifluoromethyl groups is estimated from the entropy computed from molecular data. A reasonable but not an entirely satisfactory interpretation of the heat capacity of the gas is also obtained with a barrier of the above order of magnitude.

INTRODUCTION

Octafluoropropane is an interesting molecule because of the presence of two trifluoromethyl groups which would be expected to be restricted in their rotation about their threefold axes. Previous attempts to obtain information on the rotational barrier have failed because of the incomplete experimental data. No microwave data exist for this molecule. The infrared spectrum of gaseous octafluoropropane between 4000 and 400 cm^{-1} and the Raman spectrum of the liquid have been reported by Edgell *et al.*¹ These investigators made only a partial assignment of the frequencies of the normal modes of vibrations. However, the spectral

information was insufficient to correlate the potential barrier with the experimental gaseous heat-capacity data which was later measured by Masi and co-workers² in the range 283.15° to 363.15°K.

Our approach to the problem has been made from two directions. By repeating some of the spectral work and extending the infrared observations to the region 400 to 35 cm^{-1} , we have been able to make a reasonable assignment for all the frequencies of the normal vibration with the exception of the two modes of internal rotation.³ In addition, a low-temperature calorimetric investigation has been carried out with the purpose of determining the entropy of the gas.

¹ W. F. Edgell, H. D. Mallory, and D. G. Weiblen, *J. Am. Chem. Soc.* **72**, 4856 (1950).

² J. F. Masi, H. W. Flieger, Jr., and J. S. Wicklund, *J. Res. Natl. Bur. Std.* **52**, 275 (1954).

³ E. L. Pace, A. C. Plaus, and H. V. Samuelson, *Spectrochim. Acta* **22**, 993 (1966).